

IFG: Internet-Scale Guidance for Functional Grasping Generation

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VLMs

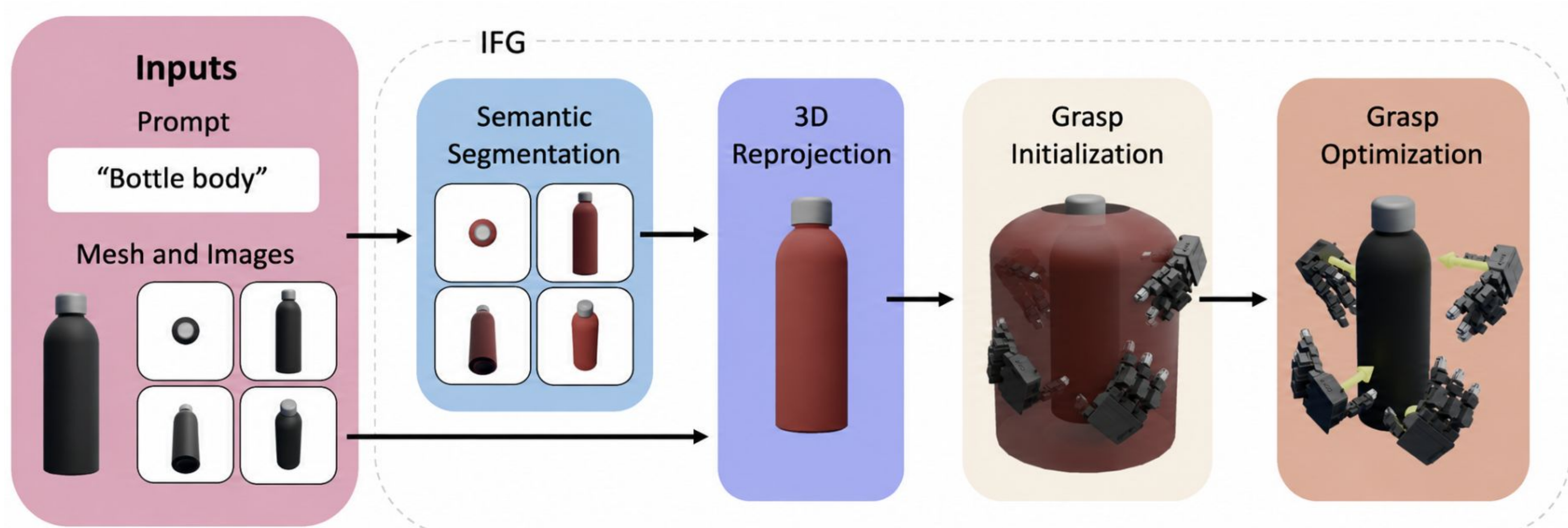
can identify grasp affordance
but lacks geometric
understanding

Simulation

can generate grasps in mass
but is not affordance aware

How to get the best of both worlds?

IFG is a functional grasp generation pipeline that synthesizes millions of force-closure grasps in simulation.



Robust on Challenging Objects

Object	DEXGRASPNET2	GRASPTTA	ISAGRASP	OURS
<i>Selected Individual Objects</i>				
Tomato Soup Can	47.8	38.3	52.0	45.5
Mug	33.2	26.9	22.6	60.4
Drill	32.1	20.8	36.4	57.5
Scissors	9.7	0.0	33.7	20.2
Screw Driver	0.0	8.3	40.0	22.0
Shampoo Bottle	50.6	25.4	18.8	53.1
Elephant Figure	23.6	29.6	24.2	35.8
Peach Can	61.8	28.0	55.3	60.3
Face Cream Tube	32.1	22.5	20.7	35.5
Tape Roll	22.7	13.9	9.8	43.2
Camel Toy	12.8	14.3	21.3	21.8
Body Wash	40.2	22.3	29.4	58.3
Object Average	30.55	20.86	30.35	42.80
Scene Average	36.71	25.64	32.51	34.16

Crowded-scene lift success rates. Both success rates of grasps on selected challenging objects and of all grasps across all scenes are reported. IFG significantly outperforms baselines on difficult objects and has comparable performance on scene average success.

1) Lift VLM segmentation to 3D region

IFG takes an object mesh and a task prompt as input, applies a VLM-based segmentation model and reprojects the results into 3D space to identify task-relevant regions.



2) Force Closure Grasp Synthesis

For geometric grounding, IFG initializes prototypes at the regions proposed by the VLM and optimizes for functional grasps using the energy function below that maximizes force closure, hand-object distance and penalizes joint limit and penetrations.

$$E = E_{fc} + w_{dis}E_{dis} + w_{joints}E_{joints} + w_{pen}E_{pen} + w_{spen}E_{spen}$$



3) Diffusion Sampler Distillation

Because this pipeline is slow and requires ground-truth observations, the generated dataset is distilled into a diffusion model that can operate on camera point clouds.

